

Model B Comparison: Original vs Current Design

METIMMOX-1 Economic Evaluation

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1 Introduction

This document compares the **Original** and **Current** Model B (Focused) designs to document the methodological change and its impact on survival predictions.

1.1 Summary of Changes

Aspect	Original Model B	Current Model B
Biomarkers as main effects	crp + tlr + tmb_braf (all three)	crp + tmb_braf (tlr dropped as main effect)
Interaction terms	One per strategy (crp:Rx, tlr:Rx, or tmb_braf:Rx)	CRP/TMB: both (crp:Rx + tmb_braf:Rx); TLR: tlr:Rx only
Strategies evaluated	control, crp, tlr, tmb_braf	control, crp, tlr, tmb_braf
Number of distinct formulas	3 (one per strategy, all unique)	2 (CRP/TMB shared + TLR separate)

1.2 Rationale for Change

The Original Model B included all three biomarkers (crp, tlr, tmb_braf) as main effects in every strategy formula, with only the strategy-specific biomarker having a treatment interaction. The Current Model B was redesigned to:

1. **Drop tlr as a main effect** - TLR is not included as a prognostic main effect in other strategies' formulas
2. **Add both CRP and TMB/BRAF interactions** for the CRP and TMB/BRAF strategies - these two biomarkers share a formula with both interaction terms (**crp:Rx + tmb_braf:Rx**)
3. **Keep TLR interaction only** for the TLR strategy - TLR's treatment effect is estimated with only **tlr:Rx**, adjusting for CRP and TMB/BRAF as main effects
4. **Use centralized configuration** via `model_configs.R` as the single source of truth

2 Methodology

2.1 Formula Definitions

2.1.1 Original Model B (per-strategy, all 3 biomarkers as main effects)

Each strategy has its own formula with all three biomarkers as main effects plus only the strategy-specific interaction:

- CRP: $\text{Surv}(\dots) \sim \text{Age} + \text{sex} + \text{Rx} + \text{crp} + \text{tlr} + \text{tmb_braf} + \text{crp:Rx}$
- TLR: $\text{Surv}(\dots) \sim \text{Age} + \text{sex} + \text{Rx} + \text{crp} + \text{tlr} + \text{tmb_braf} + \text{tlr:Rx}$
- TMB/BRAF: $\text{Surv}(\dots) \sim \text{Age} + \text{sex} + \text{Rx} + \text{crp} + \text{tlr} + \text{tmb_braf} + \text{tmb_braf:Rx}$

2.1.2 Current Model B (per-strategy, CRP/TMB shared + TLR separate)

CRP and TMB/BRAF share a formula with both interactions; TLR has its own formula:

- CRP: $\text{Surv}(\dots) \sim \text{Age} + \text{sex} + \text{Rx} + \text{crp} + \text{tmb_braf} + \text{crp:Rx} + \text{tmb_braf:Rx}$
- TLR: $\text{Surv}(\dots) \sim \text{Age} + \text{sex} + \text{Rx} + \text{crp} + \text{tmb_braf} + \text{tlr:Rx}$
- TMB/BRAF: $\text{Surv}(\dots) \sim \text{Age} + \text{sex} + \text{Rx} + \text{crp} + \text{tmb_braf} + \text{crp:Rx} + \text{tmb_braf:Rx}$

2.2 Formula Comparison Table

Table 2: Formula Comparison: Original vs Current Model B. All formulas share the common prefix $\text{Age} + \text{sex} + \text{Rx}$. OS and PFS use the same right-hand side.

Strategy	Original Model B		Current Model B	
	Main Effects	Interactions	Main Effects	Interactions
CRP	$\text{crp} + \text{tlr} + \text{tmb/braf}$	crp:Rx	$\text{crp} + \text{tmb/braf}$	$\text{crp:Rx} + \text{tmb/braf:Rx}$
TLR	$\text{crp} + \text{tlr} + \text{tmb/braf}$	tlr:Rx	$\text{crp} + \text{tlr} + \text{tmb/braf}$	tlr:Rx
TMB/BRAF	$\text{crp} + \text{tlr} + \text{tmb/braf}$	tmb/braf:Rx	$\text{crp} + \text{tmb/braf}$	$\text{crp:Rx} + \text{tmb/braf:Rx}$

Key differences:

- **Main effects:** Original includes `tlr` as a main effect in all formulas; Current drops `tlr`
- **Interactions:** Original has one interaction per strategy; Current CRP/TMB have *both* $\text{crp:Rx} + \text{tmb_braf:Rx}$
- **TLR formula:** Original adjusts for all 3 biomarkers ($\text{crp} + \text{tlr} + \text{tmb_braf}$); Current adjusts for $\text{crp} + \text{tmb_braf}$ only

3 Model Fitting

3.1 Model Fit Comparison (AIC)

Table 3: AIC Comparison: Original vs Current Model B

Model	OS AIC	PFS AIC
Original B-CRP	620.1	451.1
Original B-TLR	618.7	452.2
Original B-TMB_BRAF	620.1	451.7
Current B-CRP	630.7	481.8
Current B-TLR	618.7	452.2
Current B-TMB_BRAF	630.7	481.8

4 Parameter Estimates Comparison

4.1 CRP Strategy - OS Coefficients

Table 4: Original Model B-CRP - OS Coefficients

Term	Estimate	SE	HR (95% CI)
shape	1.201	0.182	3.325 (2.329-4.747)
rate	-3.179	0.589	0.042 (0.013-0.132)
Age	0.003	0.008	1.003 (0.987-1.019)
sex1	0.167	0.149	1.182 (0.883-1.582)
RxExperimental arm	-0.046	0.194	0.955 (0.653-1.398)
crp1	-0.429	0.314	0.651 (0.352-1.206)
tlr1	-0.573	0.168	0.564 (0.406-0.784)
tmb_braf1	-0.139	0.154	0.870 (0.644-1.176)
RxExperimental arm:crp1	0.136	0.380	1.146 (0.544-2.414)

Table 5: Current Model B-CRP - OS Coefficients

Term	Estimate	SE	HR (95% CI)
shape	1.058	0.181	2.881 (2.020-4.108)
rate	-3.605	0.623	0.027 (0.008-0.092)
Age	-0.001	0.009	0.999 (0.983-1.016)
sex1	0.202	0.158	1.224 (0.898-1.670)
RxExperimental arm	0.261	0.224	1.298 (0.836-2.016)
crp1	-0.556	0.338	0.574 (0.296-1.113)
tmb_braf1	-0.042	0.239	0.959 (0.600-1.532)
RxExperimental arm:crp1	0.029	0.408	1.029 (0.463-2.290)
RxExperimental arm:tmb_braf1	-0.111	0.325	0.895 (0.474-1.691)

4.2 TLR Strategy - OS Coefficients

Table 6: Original Model B-TLR - OS Coefficients

Term	Estimate	SE	HR (95% CI)
shape	1.226	0.182	3.408 (2.386-4.869)
rate	-2.896	0.611	0.055 (0.017-0.183)
Age	0.002	0.008	1.002 (0.986-1.018)
sex1	0.145	0.148	1.156 (0.864-1.545)
RxExperimental arm	-0.268	0.256	0.765 (0.463-1.263)
crp1	-0.361	0.177	0.697 (0.493-0.986)
tlr1	-0.808	0.247	0.446 (0.275-0.722)
tmb_braf1	-0.139	0.148	0.870 (0.651-1.163)
RxExperimental arm:tlr1	0.402	0.316	1.495 (0.805-2.775)

Table 7: Current Model B-TLR - OS Coefficients

Term	Estimate	SE	HR (95% CI)
shape	1.226	0.182	3.408 (2.385-4.868)
rate	-2.895	0.611	0.055 (0.017-0.183)
Age	0.002	0.008	1.002 (0.986-1.018)
sex1	0.145	0.148	1.156 (0.864-1.545)
RxExperimental arm	-0.269	0.256	0.764 (0.463-1.262)
crp1	-0.361	0.177	0.697 (0.493-0.986)

tmb_braf1	-0.139	0.148	0.870 (0.651-1.162)
RxControl arm:tlr1	-0.809	0.247	0.445 (0.275-0.722)
RxExperimental arm:tlr1	-0.407	0.208	0.666 (0.443-1.000)

4.3 TMB/BRAF Strategy - OS Coefficients

Table 8: Original Model B-TMB/BRAF - OS Coefficients

Term	Estimate	SE	HR (95% CI)
shape	1.205	0.182	3.335 (2.336-4.761)
rate	-3.243	0.563	0.039 (0.013-0.118)
Age	0.003	0.008	1.003 (0.988-1.019)
sex1	0.170	0.148	1.185 (0.886-1.585)
RxExperimental arm	0.032	0.202	1.033 (0.696-1.534)
crp1	-0.319	0.182	0.727 (0.509-1.038)
tlr1	-0.569	0.167	0.566 (0.408-0.786)
tmb_braf1	-0.062	0.221	0.940 (0.609-1.449)
RxExperimental arm:tmb_braf1	-0.117	0.304	0.890 (0.491-1.614)

Table 9: Current Model B-TMB/BRAF - OS Coefficients

Term	Estimate	SE	HR (95% CI)
shape	1.058	0.181	2.881 (2.020-4.108)
rate	-3.605	0.623	0.027 (0.008-0.092)
Age	-0.001	0.009	0.999 (0.983-1.016)
sex1	0.202	0.158	1.224 (0.898-1.670)
RxExperimental arm	0.261	0.224	1.298 (0.836-2.016)
crp1	-0.556	0.338	0.574 (0.296-1.113)
tmb_braf1	-0.042	0.239	0.959 (0.600-1.532)
RxExperimental arm:crp1	0.029	0.408	1.029 (0.463-2.290)
RxExperimental arm:tmb_braf1	-0.111	0.325	0.895 (0.474-1.691)

4.4 CRP Strategy - PFS Coefficients

Table 10: Original Model B-CRP - PFS Coefficients

Term	Estimate	SE	HR (95% CI)
shape	1.215	0.190	3.370 (2.324-4.888)
rate	-1.680	0.597	0.186 (0.058-0.600)
Age	-0.001	0.009	0.999 (0.982-1.016)
sex1	0.115	0.165	1.122 (0.813-1.550)
RxExperimental arm	0.169	0.197	1.184 (0.805-1.742)
crp1	0.014	0.420	1.014 (0.445-2.311)
tlr1	-1.114	0.168	0.328 (0.236-0.456)
tmb_braf1	-0.323	0.174	0.724 (0.514-1.018)
RxExperimental arm:crp1	-0.704	0.485	0.494 (0.191-1.280)

Table 11: Current Model B-CRP - PFS Coefficients

Term	Estimate	SE	HR (95% CI)
shape	0.707	0.181	2.027 (1.421-2.893)
rate	-2.507	0.773	0.082 (0.018-0.371)
Age	-0.011	0.011	0.989 (0.968-1.011)

sex1	0.139	0.210	1.149 (0.762-1.734)
RxExperimental arm	0.560	0.296	1.751 (0.980-3.131)
crp1	-0.263	0.542	0.769 (0.266-2.222)
tmb_braf1	-0.178	0.337	0.837 (0.432-1.622)
RxExperimental arm:crp1	-0.661	0.620	0.516 (0.153-1.741)
RxExperimental arm:tmb_braf1	-0.389	0.449	0.678 (0.281-1.634)

4.5 TLR Strategy - PFS Coefficients

Table 12: Original Model B-TLR - PFS Coefficients

Term	Estimate	SE	HR (95% CI)
shape	1.194	0.190	3.299 (2.273-4.788)
rate	-1.711	0.634	0.181 (0.052-0.626)
Age	-0.002	0.009	0.998 (0.981-1.015)
sex1	0.128	0.170	1.137 (0.815-1.585)
RxExperimental arm	0.237	0.285	1.267 (0.725-2.215)
crp1	-0.527	0.206	0.590 (0.394-0.885)
tlr1	-0.928	0.273	0.395 (0.232-0.675)
tmb_braf1	-0.387	0.169	0.679 (0.488-0.947)
RxExperimental arm:tlr1	-0.295	0.349	0.745 (0.376-1.476)

Table 13: Current Model B-TLR - PFS Coefficients

Term	Estimate	SE	HR (95% CI)
shape	1.196	0.190	3.305 (2.278-4.796)
rate	-1.712	0.634	0.180 (0.052-0.625)
Age	-0.002	0.009	0.998 (0.981-1.015)
sex1	0.127	0.169	1.136 (0.815-1.582)
RxExperimental arm	0.238	0.285	1.269 (0.726-2.217)
crp1	-0.527	0.206	0.590 (0.394-0.884)
tmb_braf1	-0.385	0.169	0.680 (0.488-0.947)
RxControl arm:tlr1	-0.928	0.273	0.395 (0.232-0.675)
RxExperimental arm:tlr1	-1.224	0.218	0.294 (0.192-0.451)

4.6 TMB/BRAF Strategy - PFS Coefficients

Table 14: Original Model B-TMB/BRAF - PFS Coefficients

Term	Estimate	SE	HR (95% CI)
shape	1.197	0.190	3.311 (2.283-4.804)
rate	-1.556	0.576	0.211 (0.068-0.653)
Age	-0.003	0.008	0.997 (0.980-1.013)
sex1	0.097	0.167	1.102 (0.795-1.528)
RxExperimental arm	0.181	0.218	1.198 (0.782-1.837)
crp1	-0.458	0.222	0.633 (0.409-0.978)
tlr1	-1.105	0.168	0.331 (0.238-0.460)
tmb_braf1	-0.191	0.250	0.826 (0.506-1.349)
RxExperimental arm:tmb_braf1	-0.388	0.347	0.678 (0.343-1.340)

Table 15: Current Model B-TMB/BRAF - PFS Coefficients

Term	Estimate	SE	HR (95% CI)
shape	0.707	0.181	2.027 (1.421-2.893)
rate	-2.507	0.773	0.082 (0.018-0.371)
Age	-0.011	0.011	0.989 (0.968-1.011)
sex1	0.139	0.210	1.149 (0.762-1.734)
RxExperimental arm	0.560	0.296	1.751 (0.980-3.131)
crp1	-0.263	0.542	0.769 (0.266-2.222)
tmb_braf1	-0.178	0.337	0.837 (0.432-1.622)
RxExperimental arm:crp1	-0.661	0.620	0.516 (0.153-1.741)
RxExperimental arm:tmb_braf1	-0.389	0.449	0.678 (0.281-1.634)

5 Survival Curves Comparison

5.1 Overall Survival Curves

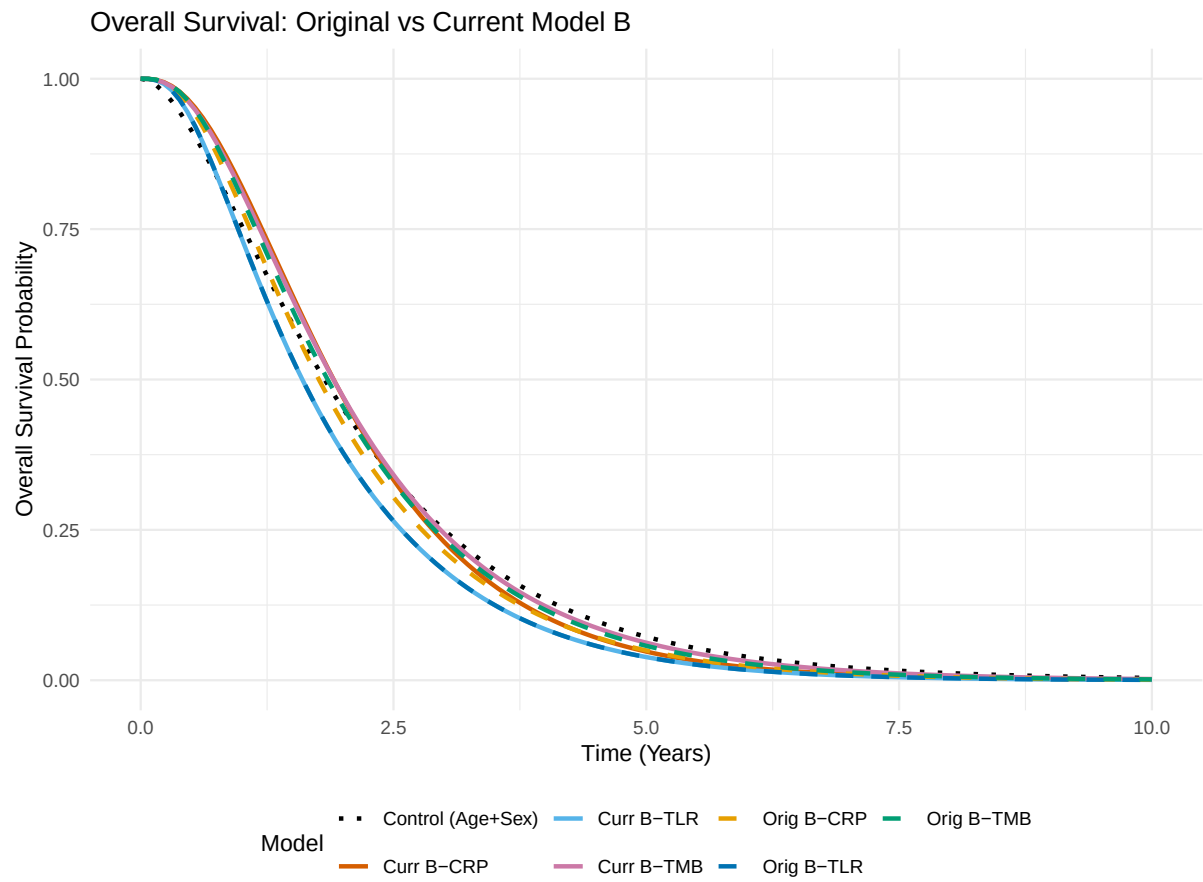


Figure 1: Overall Survival Curves: Original vs Current Model B (population-weighted strategies + control reference)

5.2 Progression-Free Survival Curves

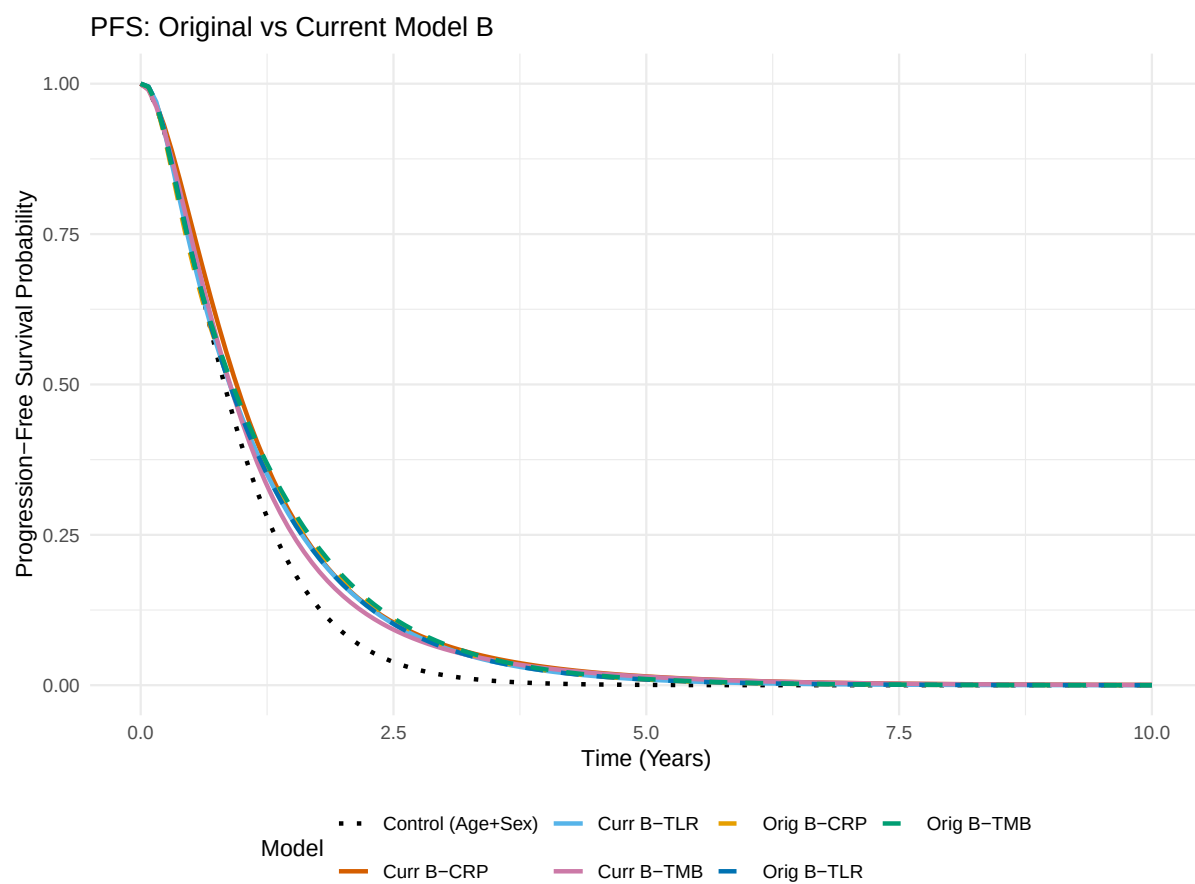


Figure 2: Progression-Free Survival Curves: Original vs Current Model B (population-weighted strategies + control reference)

6 Life Years, QALYs, and Costs

This section compares discounted and undiscounted life-years, QALYs, costs, and ICERs between Original and Current Model B specifications.

6.1 Life-Years Comparison

Life-years are calculated as the area under the survival curve (restricted mean survival time) over the 10-year time horizon. Population-weighted estimates for each biomarker strategy. Discounting at 4% per year.

Table 16: Life-Years Comparison: Original vs Current Model B (population-weighted)

Outcome	Control	CRP			TLR			TMB/BRAF		
		Old	New	Delta	Old	New	Delta	Old	New	Delta
Discounted LY (OS)	2.106	2.011	2.097	+0.086	1.859	1.858	-0.001	2.097	2.148	+0.051
Discounted LY (PFS)	0.969	1.165	1.208	+0.043	1.153	1.152	-0.001	1.184	1.146	-0.038
Undiscounted LY (OS)	2.252	2.134	2.226	+0.091	1.965	1.964	-0.001	2.231	2.288	+0.058
Undiscounted LY (PFS)	0.997	1.213	1.261	+0.047	1.200	1.199	-0.001	1.234	1.194	-0.040

6.2 QALYs Comparison

QALYs are computed using weekly-resolution OS and PFS predictions, with utility weights of 0.9077 (progression-free) and 0.9005 (progressed), discounted at 4% per year.

Table 17: QALYs Comparison: Original vs Current Model B (population-weighted, discounted)

Outcome	Control	CRP			TLR			TMB/BRAF		
		Old	New	Delta	Old	New	Delta	Old	New	Delta
QALYs	1.9037	1.8193	1.8970	+0.0777	1.6820	1.6812	-0.0008	1.8968	1.9422	+0.0454

6.3 Cost Comparison

Costs include drug costs (nivolumab, FLOX), test costs (CT, blood, NGS), visit costs, follow-up costs, and end-of-life costs. Biomarker-positive patients receive experimental treatment; biomarker-negative patients receive standard treatment. Costs are prevalence-weighted and discounted.

Table 18: Discounted Costs: Original vs Current Model B

Strategy	Original B (Cost)	Current B (Cost)	Difference (EUR)
Control	EUR 21,212	EUR 21,212	-
CRP	EUR 55,114	EUR 54,083	-1031
TLR	EUR 83,761	EUR 83,767	+6
TMB/BRAF	EUR 64,376	EUR 63,373	-1003

6.4 ICER Comparison - Original Model B

Incremental cost-effectiveness ratios (ICERs) for the Original Model B specification, comparing each biomarker strategy to the control (standard of care).

Table 19: Cost-Effectiveness: Original Model B (vs Control)

Strategy	Cost	QALY	Inc. Cost	Inc. QALY	ICER
Control	EUR 21,212	1.9037	-	-	Reference
CRP	EUR 55,114	1.8193	EUR 33,902	-0.0845	Dominated
TLR	EUR 83,761	1.6820	EUR 62,549	-0.2217	Dominated
TMB/BRAF	EUR 64,376	1.8968	EUR 43,164	-0.0069	Dominated

6.5 ICER Comparison - Current Model B

Table 20: Cost-Effectiveness: Current Model B (vs Control)

Strategy	Cost	QALY	Inc. Cost	Inc. QALY	ICER
Control	EUR 21,212	1.9037	-	-	Reference
CRP	EUR 54,083	1.8970	EUR 32,871	-0.0068	Dominated
TLR	EUR 83,767	1.6812	EUR 62,554	-0.2225	Dominated
TMB/BRAF	EUR 63,373	1.9422	EUR 42,161	0.0385	EUR 1,095,026

6.6 ICER Change: Original vs Current Model B

Table 21: ICER Change from Original to Current Model B Specification

Strategy	Original ICER	Current ICER	Change (EUR)	Change (%)
CRP	Dominated	Dominated	-	-
TLR	Dominated	Dominated	-	-
TMB/BRAF	Dominated	EUR 1,095,026	-	-

7 Summary

The redesign of Model B from the Original specification to the Current specification involves two key structural changes:

7.1 Key Differences

1. **tlr dropped as a main effect:** The Original Model B included all three biomarkers (crp, tlr, tmb_braf) as main effects in every strategy formula, adjusting for each biomarker's prognostic effect. The Current Model B only includes crp and tmb_braf as main effects, dropping tlr.
2. **Interaction structure changed:** The Original Model B had exactly one treatment interaction per strategy (the strategy's own biomarker). The Current Model B uses both `crp:Rx + tmb_braf:Rx` for the CRP and TMB/BRAF strategies, while TLR uses only `tlr:Rx`.
3. **Both versions include all three strategies:** Both Original and Current Model B evaluate all four strategies (control, CRP, TLR, TMB/BRAF) with per-strategy formulas.

7.2 Rationale

The Current Model B reflects the analytical judgment that:

- CRP and TMB/BRAF treatment effects are best estimated jointly (shared formula with both interactions)
- TLR's treatment effect should be estimated with its own interaction term only
- TLR's prognostic effect (as a main effect) is not needed for adjustment in other formulas

7.3 Impact on Results

The comprehensive economic comparison above quantifies the impact of these formula changes across multiple dimensions:

- **Parameter estimates:** OS and PFS coefficients for all three biomarker strategies, showing how coefficient values change between specifications
- **Survival:** Population-weighted OS and PFS curves comparing predicted survival trajectories
- **Life-years:** Both discounted and undiscounted restricted mean survival times
- **QALYs:** Quality-adjusted outcomes incorporating utilities for progression-free and progressed states
- **Costs:** Full strategy-level costs including drug, test, visit, follow-up, and end-of-life costs
- **ICERs:** Incremental cost-effectiveness ratios versus control, and the change in ICERs between specifications

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